

Notes on the Likelihood Function

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Abstract

Notes developing the likelihood function from the binomial example: the shape of $L(q \mid v, n)$, the sense in which the likelihood “is” the probability mass function with argument and parameter exchanged, and the measure-theoretic definition together with the three qualifications it carries.

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1 The binomial example and the shape of the likelihood

Consider the binomial model with probability of success q and n known. The probability mass function of the count Υ is

$$f_{\Upsilon}(v; q, n) = \mathbb{P}(\Upsilon = v \mid n, q) = \binom{n}{v} q^v (1 - q)^{n-v}, \quad v \in \{0, 1, \dots, n\}. \quad (1)$$

Fixing an observed count v and reading (1) as a function of q gives the likelihood

$$L(q \mid v, n) = \binom{n}{v} q^v (1 - q)^{n-v}, \quad q \in [0, 1]. \quad (2)$$

Why it has that shape. As a function of q , expression (2) is the kernel of a $\text{Beta}(v+1, n-v+1)$ density; everything about its shape follows from that. The binomial coefficient is constant in q : it rescales vertically and plays no role in location or shape. The shape is the product of two competing power terms — q^v rewards larger q (pulling mass toward 1), while $(1 - q)^{n-v}$ rewards smaller q (pulling toward 0) — and they balance at an interior point.

Work with the log-likelihood

$$\ell(q) = \text{const} + v \log q + (n - v) \log(1 - q). \quad (3)$$

Then

$$\ell'(q) = \frac{v}{q} - \frac{n - v}{1 - q} = 0 \implies v(1 - q) = (n - v)q \implies q = \frac{v}{n}, \quad (4)$$

which is exactly the MLE, and

$$\ell''(q) = -\frac{v}{q^2} - \frac{n-v}{(1-q)^2} < 0 \quad \text{on } (0, 1), \quad (5)$$

so ℓ is strictly concave. Hence L is strictly log-concave, therefore unimodal with a single peak at $q = v/n$.

For $0 < v < n$ the endpoints vanish, $L(0) = L(1) = 0$, giving the characteristic single bump: rising from 0, peaking at v/n , decaying back to 0. The degenerate cases collapse the bump against a wall: $v = 0$ gives $(1-q)^n$, monotone decreasing with mode at $q = 0$; $v = n$ gives q^n , monotone increasing with mode at $q = 1$.

Remark. L is smooth (a degree- n polynomial in q), which is what is meant by calling it “continuous.” It is *not* a density in q : it does not integrate to 1 over q . Normalizing requires dividing by $B(v+1, n-v+1)$, and the fact that this normalization produces the Beta distribution is precisely Beta–Binomial conjugacy.

2 PMF and likelihood as slices of one surface

The likelihood is not *computed* from the PMF by applying an operation; it *is* the PMF, with the roles of argument and parameter exchanged. “Same functional form” is not a derivation and not a coincidence — it is the definition of the likelihood function.

Fix n and consider the single bivariate function

$$g(v, q) = \binom{n}{v} q^v (1-q)^{n-v}, \quad (v, q) \in \{0, 1, \dots, n\} \times [0, 1]. \quad (6)$$

The PMF and the likelihood are two orthogonal slices of this one surface:

- Fix q , let v vary over $\{0, \dots, n\}$: this is the PMF $f_{\mathbf{r}}(v; q, n)$, a slice along the discrete axis.
- Fix the observed v , let q vary over $[0, 1]$: this is $L(q | v, n)$, a slice along the continuous axis.

So the “computation” is only: substitute the observed v and read what remains as a function of q . The binomial coefficient and both power terms stay algebraically identical, which is why the two expressions are typographically the same. The discrete-versus-continuous distinction is entirely about which axis was left free: v ranges over a finite set, q over a continuum.

Pointwise equality is not identity of objects. The two slices normalize on different axes. Across the discrete axis the PMF sums to 1 for every q , by the binomial theorem:

$$\sum_{v=0}^n g(v, q) = (q + (1-q))^n = 1. \quad (7)$$

Across the continuous axis the likelihood integrates to

$$\int_0^1 g(v, q) dq = \binom{n}{v} B(v+1, n-v+1) = \binom{n}{v} \frac{v! (n-v)!}{(n+1)!} = \frac{1}{n+1}, \quad (8)$$

independent of v , and in particular $\neq 1$. Thus L is not a density in q ; this is exactly why maximizing it, rather than integrating it, is the sensible operation, and why turning it into a density requires the Beta normalization.

Remark. That the integral (8) equals $\frac{1}{n+1}$ for every v is the statement that a uniform prior on q induces a uniform prior-predictive on the count $v \in \{0, \dots, n\}$ — the Beta(1, 1)–Binomial edge case.

3 The likelihood function, formally

Definition 3.1 (Likelihood function). Let $\{P_\theta : \theta \in \Theta\}$ be a statistical model, a family of probability measures on a sample space $(\mathcal{X}, \mathcal{A})$, all dominated by a common σ -finite measure μ (counting measure in the discrete case, Lebesgue measure in the continuous case). By the Radon–Nikodym theorem each P_θ has a density

$$f(x | \theta) = \frac{dP_\theta}{d\mu}(x). \quad (9)$$

Given an observed outcome x , the *likelihood function* is the map

$$L(\cdot | x) : \Theta \rightarrow [0, \infty), \quad L(\theta | x) = f(x | \theta). \quad (10)$$

The single content-bearing move is the exchange of roles from Section 2: it is the same function $f(x | \theta)$, but x is held fixed (the data observed) and θ is the free variable. The likelihood is a function on the parameter space, not on the sample space.

Three points that the informal “density read backwards” phrasing obscures:

1. **Defined only up to a θ -independent positive factor.** The Radon–Nikodym derivative depends on the dominating measure: switching $\mu \rightarrow \tilde{\mu}$ with $\frac{d\mu}{d\tilde{\mu}} = h(x)$ multiplies every density, hence $L(\theta | x)$, by the constant-in- θ factor $h(x)$. This changes nothing inferential — the $\arg \max$, the likelihood ratios $L(\theta_1 | x)/L(\theta_2 | x)$, and the score $\nabla_\theta \log L$ are all invariant. The likelihood is properly an equivalence class under multiplication by positive functions of x alone. This is why the $\binom{n}{v}$ in Section 1 was inert.
2. **$L(\theta | x)$ is not a density in θ .** There is no reference measure on Θ in the definition, and $\int_\Theta L(\theta | x) d\theta$ need not be finite, let alone 1 (the $\frac{1}{n+1}$ computation of (8)). The likelihood becomes a probabilistic object on Θ only after a prior is supplied and the product normalized (Bayes).
3. **The fixed x must lie in the support.** Where all P_θ assign zero μ -density the derivative is defined only μ -a.e., so L is genuinely ambiguous there — harmless, since one evaluates at observed data.

The iid case. For iid data $x = (x_1, \dots, x_m)$ from $f(\cdot | \theta)$, the product structure of the joint density gives

$$L(\theta | x) = \prod_{i=1}^m f(x_i | \theta), \quad (11)$$

and one works with the log-likelihood

$$\ell(\theta) = \log L(\theta | x) = \sum_{i=1}^m \log f(x_i | \theta), \quad (12)$$

the MLE being any $\hat{\theta} \in \arg \max_{\theta \in \Theta} \ell(\theta)$.